You are here: Applications and Workflows > QikProp > User Manual > QikProp Accuracy Issues

QikProp Accuracy Issues

QikProp predictions are performed for specific conditions (pH 7, for one), and comparison of the results with experiment must take into account any differences in those conditions. The following issues can affect the accuracy of the comparison:

- active vs passive transport mechanisms
- efflux pump mechanisms
- smaller number of compounds in the training sets
- different experimental conditions than that used in the regression
- noise in experimental data due to inherent complexity of chemical systems interacting with biological systems

The biggest discrepancies are for PCaco predictions, owing to these issues. Values for log BB are less problematic. The QikProp fit for log BB is very good—at the noise level of the data—but there are a few outliers such as oxazepam and tiotidine that are not fitted by any regression method.

Obtaining reliable experimental data for log S requires pure, non-hydrated, no-salt crystals in equilibrium with a saturated aqueous solution. This condition is often difficult to achieve, so it is important to check the experiments as well as the QikProp predictions.

Comparisons must take into account the protonation state of molecules at physiological pH. For example, compounds that contain a tertiary amine are protonated at that pH, and log P measurements must be performed at pH 11, and corrected to pH 7.

Experimental log P values (such as log Po/w) for amino acids, which have a saturated amine and a carboxylic acid, are affected by the protonation state. Though the Hansch * value is supposed to correspond to the un-ionized state, there is no pH at which either the amine is not protonated or the COOH deprotonated or both. The Hansch * values in these cases correspond to a pseudo log D at pH 7, at which pH the acid group is ionized, but the amino group’s protonation has been corrected for. The data in Table 1 clearly show the problem.

*Table 1. Hansch * values for selected molecules*

Molecule	* Value	Molecule	* Value
ethanol HOCH2CH3	−0.31	hydroxyacetic acid HOCH2COOH	−1.11
ethylamine H2NCH2CH3	−0.13	aminoacetic acid (glycine) H2NCH2COOH	−3.21

Molecule	* Value	Molecule	* Value
difference	−0.18	difference	+2.10

The difference for an alcohol and an un-ionized amine should be −0.18, but for the two molecules that have an acid group, the difference is +2.10. If the acid in both cases was un-ionized, the difference should still be −0.18. In fact, the value of −1.11 is for the un-ionized acid, but the value of −3.21 is for the ionized acid. The discrepancy of 2.28 is exactly what is expected for the un-ionized acid, i.e.

pH 7.0 − pKa of RCOOH (4.8) = 2.2

The true un-ionized Hansch * value for glycine should be about −1.0; QikProp 1.5 gives −0.96.

The above issues do not apply to peptides in general, which may be capped. QikProp 1.5 agrees with Hansch * values for capped peptides, as illustrated in Table 2.

*Table 2. Comparison of Hansch * values and QikProp predictions for capped peptides*

Peptide	Hansch * Value
AcMetValNH2	
Ac(Ala)3NHtBu	
AcAlaTyrLeuNH2	

Therefore, for QikProp versions since 1.6, the log Po/w values for amino acids have been lowered by 2.0 log units, so that QikProp reproduces the Hansch * values. The correction is noted in the output files when it is applied.

Note that this correction is not just for the standard amino acids, but for any compound with a saturated amine group and a carboxylic acid, e.g., amoxicillin.

A similar correction factor is used by Hansch and Leo to account for zwitterions. They discuss this in Ref. 16, and show that it works for molecules like amoxicillin which are zwitterionic.

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